

Physics A (H156, H556)

Exploring - Charge ppq KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: Not set

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is **32**.
- The total number of marks may take into account some 'either/or' question choices.

1(a) A gamma camera has several important components including a collimator, scintillator and photomultiplier tubes.

Suggest why the collimator needs to be long and narrow.

----- [1]

(b) State the function of the scintillator.

----- [1]

(c) In a single photomultiplier tube, a photon of light produces a $0.32\ \mu\text{A}$ pulse of current for a duration of $1.2\ \text{ns}$.

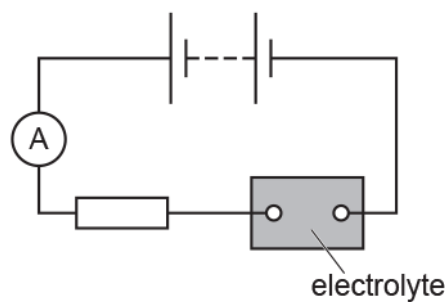
Calculate the number of electrons responsible for this pulse of current.

number of electrons = [2]

(d) State one diagnostic application of a gamma camera.

----- [1]

2 A current is present in the circuit below.



The resistor is made from a length of wire.

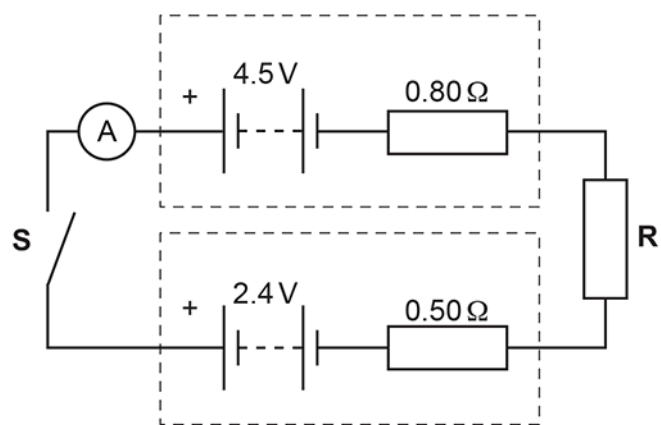
Which row gives the correct charge carriers in the resistor and in the electrolyte?

	Charge carriers in the resistor	Charge carriers in the electrolyte
A	Electrons	Electrons
B	Electrons	Ions
C	Electrons and protons	Ions and electrons
D	Electrons and ions	Ions and protons

Your answer

[1]

3 The circuit diagram of an electrical circuit is shown below.



The positive terminals of the batteries are connected together.
One battery has electromotive force (e.m.f.) 4.5 V and internal resistance 0.80 Ω.
The other battery has e.m.f. 2.4 V and internal resistance 0.50 Ω.
R is a coil of insulated wire of resistance 1.2 Ω at room temperature.

The switch S is closed.

i. On the diagram, draw an arrow to show the direction of the conventional current.

[1]

- ii. Calculate the current I shown by the ammeter.

$I = \dots\dots\dots$ A [3]

- iii. The insulated wire has diameter 4.6×10^{-4} m.

The number density of charge carriers in R is $4.2 \times 10^{28} \text{ m}^{-3}$

Calculate the mean drift velocity v of the charge carriers in R .

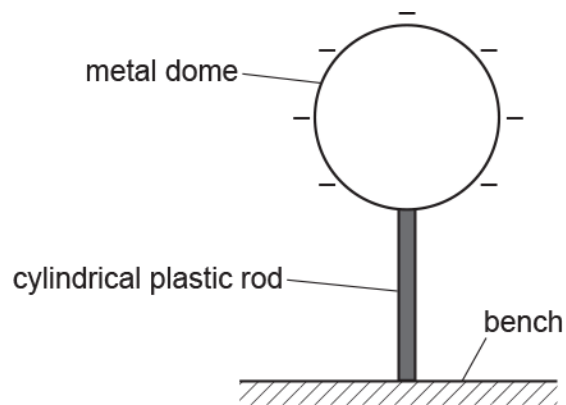
$v = \dots\dots\dots$ m s^{-1} [2]

- iv. The current measured by the ammeter is smaller than that calculated in (ii). This is because the temperature of R increased due to heating by the current.

Without any changes to the circuit itself, state and explain what practically can be done to make the measured current the same as the calculated current..

----- [2]

- 4 A spherical metal dome shown below is charged to a potential of -12 kV .



The dome is supported by a cylindrical plastic rod. The radius of the dome is 0.19 m .

- i. Show that the magnitude of the total charge Q on the dome is $2.5 \times 10^{-7}\text{ C}$.

[2]

- ii. The dome discharges slowly through the plastic rod.
It takes 78 hours for the dome to completely discharge.

- 1 Show that the mean current I in the plastic rod is about $9 \times 10^{-13}\text{ A}$.

[2]

- 2 The average potential difference across the plastic rod during discharge is 6000 V .
The rod has cross-sectional area $1.1 \times 10^{-4}\text{ m}^2$ and length 0.38 m .

Calculate the resistivity ρ of the plastic.

$\rho = \dots\dots\dots \Omega\text{m}$ [3]

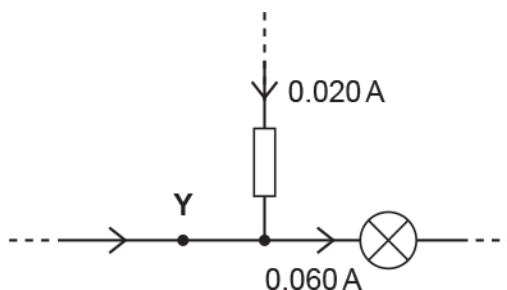
5 Which law indicates that charge is conserved?

- A Lenz's law
- B Coulomb's law
- C Kirchhoff's first law
- D Faraday's law of electromagnetic induction

Your answer

[1]

6 Part of an electric circuit is shown below.



The direction of all the currents and the magnitude of two currents are shown.

How many electrons pass through the point Y in 10 s?

- A 1.25×10^{18}
- B 2.50×10^{18}
- C 3.75×10^{18}
- D 5.00×10^{18}

Your answer

[1]

7

Derive the S.I. base units for resistance.

base units:----- [2]

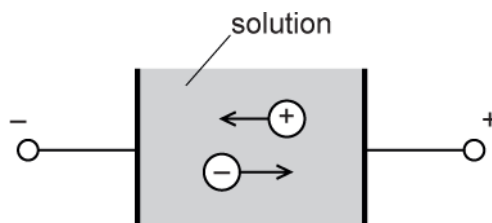
8 Which electrical quantity has S.I. units ampere-second (A s)?

- A charge
- B current
- C resistance
- D potential difference

Your answer

[1]

9 The diagram below shows the motion of positive and negative particles in a conducting solution.



Which statement is correct?

- A The current in the solution is zero.
- B The conventional current is to the left.
- C The positive particles are always protons.
- D The negative particles are always electrons.

Your answer

[1]

- 10 One million electrons travel between two points in a circuit.
The total energy gained by the electrons is 1.6×10^{-10} J.

What is the potential difference between the two points?

- A 1.6×10^{-16} V
- B 1.6×10^{-4} V
- C 1.0×10^3 V
- D 1.0×10^9 V

Your answer

[1]

- 11 A small heater is connected to a power supply. The power supply is switched on for 100 s. The current in the heater is 3.0 A and it dissipates 1200 J of thermal energy.

What is the potential difference across the heater?

- A 0.25V
- B 4.0V
- C 12V
- D 300V

Your answer

[1]

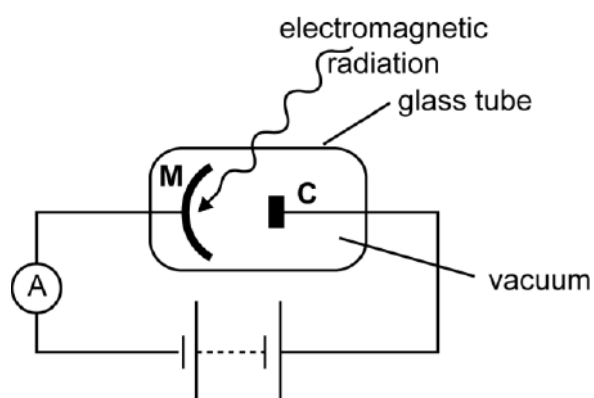


Fig. 19

When the metal **M** is exposed to electromagnetic radiation, photoelectrons are ejected from the surface of the metal. These photoelectrons are collected at the electrode **C** and the sensitive ammeter indicates the presence of a tiny current.

The work function of the metal **M** is 2.3 eV.

The incident electromagnetic radiation has wavelength 5.1×10^{-7} m.

The ammeter reading is 0.24 μ A.

Calculate the number of photoelectrons reaching **C** in a time of 5.0 s.

number = _____ [3]

END OF QUESTION PAPER

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
1	a	Allows only those gamma rays / waves / photons travelling along axis (of the collimator) to get through (and reach scintillator)	B1	Allow less fuzzy / clear image Allow all the gamma rays / waves / photons are parallel / in the same direction (to each other) Allow <u>absorbs</u> those gamma rays / waves / photons not parallel (to the axis of collimator) Allow so the photons are travelling perpendicular to the scintillator Do not allow the gamma rays are travelling vertically, unless it is clear the collimator is also vertical <u>Examiner's Comments</u> This question was well answered by around half of the candidates who appreciated that the long, narrow collimator would allow only gamma rays which were parallel to each other to be received by the scintillator. This can be expressed in a number of ways and marks were given to candidates who were able to state this using alternative descriptions. A common confusion appeared between using the terms perpendicular and parallel, with some candidates incorrectly stating that the collimator allowed photons perpendicular to the tubes to pass. Essentially, the collimator allows for a clearer image to form and so this is an acceptable response.
	b	Turn gamma (photons) into (many photons of) light	B1	Ignore reference to rays / waves Ignore reference to flash <u>Examiner's Comments</u> Around half of the candidates were able to correctly explain that the gamma photons produced visible light in the scintillator. Several candidates thought that the scintillator produced electrons or a voltage when the gamma photons were incident on it instead. Many candidates gave the extra correct information that many visible photons are produced from one gamma photon, although this was not a required detail on this occasion.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	c		number of electrons = $\frac{0.32 \times 10^{-6} \times 1.2 \times 10^{-9}}{e}$ number of electrons = 2400	C1 A1	Ignore any POT error for C1 mark <u>Examiner's Comments</u> This calculation was done well by a large majority of candidates. There were few errors in the unit prefixes, with the most common one being 1.2 ns as 1.2×10^{-12} s. A small number of candidates calculated the charge correctly, but then took it no further to determine the number of electrons.
	d		Any sensible <u>diagnostic</u> suggestion, e.g. <u>detection</u> of cancer / scans of (named) organ / scans of tissue / bone scans / observing functionality of (named) organ	B1	Not medical <u>treatment</u> e.g. radiotherapy Not body scan Ignore PET scanner Do not allow CAT scan <u>Examiner's Comments</u> The key word here was “diagnostic” and responses were expected to state any reasonable diagnostic use of a gamma camera and a wide variety of responses were given and accepted. Common responses included checking for brain tumours and observing kidney failure. Although a gamma camera may be used in a PET scanner, this alone was not acceptable as it does not explain the diagnostic use. Candidate should be encouraged to write in a clear sentence structure as simple responses such as “cancer” cannot be given marks.
			Total	5	
2			B	1	<u>Examiner's Comments</u> The vast majority of candidates were able to correctly recall the correct charge carriers in the resistor and the electrolyte. A proved to be the most common distractor, possibly as the others contained “protons”.
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
3		i	Arrow in anticlockwise direction	B1	Allow this mark for correct direction shown on diagram either on or off connecting wires <u>Examiner's Comments</u> This question required the candidates to appreciate that the sum of the emfs will lead to an anticlockwise conventional current. This question was answered well by the majority of candidates, however a number put two directions on, one from each cell.  Misconception The unusual setting out of the circuit meant that some candidates were unsure whether parts of the circuit were in series or parallel. This could have been overcome by following the circuit or even by redrawing it.
		ii	$(E =) 4.5 - 2.4$ or $(R_T =) 0.80 + 0.50 + 1.2$ $4.5 - 2.4 = I \times (0.80 + 0.50 + 1.2)$ $I = 0.84 \text{ (A)}$	C1 C1 A1	$E = 2.1 \text{ (V)}$; $R_T = 2.5 \text{ (}\Omega\text{)}$ Treat missing 1.2 resistance as TE Allow 2 marks for 2.8 (A); $E = 6.9 \text{ V}$ used <u>Examiner's Comments</u> This calculation required the candidate to set out the whole circuit in one. Around one third did not score any marks on this question as they attempted to treat each cell individually and then produce some form of average. Other common misunderstandings included treating the 0.5 ohm and 0.8 ohm resistors as if they were in parallel, and adding the emfs.

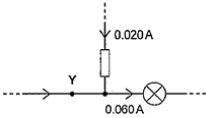
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iii	$(I = Anev)$ $0.84 = \pi \times (2.3 \times 10^{-4})^2 \times 4.2 \times 10^{28} \times 1.60 \times 10^{-19} \times v$ $v = 7.5 \times 10^{-4} \text{ (m s}^{-1}\text{)}$	C1 A1	Possible ECF from (ii) Note answer is $2.5 \times 10^{-3} \text{ (m s}^{-1}\text{)}$ for $I = 2.76 \text{ (A)}$ Allow 1 mark for 1.9×10^{-4} ; diameter used as radius <u>Examiner's Comments</u> This question was well done by a large number of candidates, many of whom scored full marks by correctly following through with their value of current from the previous part. Few candidates used the diameter instead of the radius when determining the cross sectional area, and for the most part the setting out of the calculation meant that credit could be given even if arithmetic errors occurred later.
		iv	Sensible suggestion, e.g. use a water bath / fan / only switch on when taking readings Need to lower the temperature / reduce resistance of R	M1 A1	Allow keep the surroundings cold Allow to keep the temperature / resistance constant OR allow increase in temperature increases resistance <u>Examiner's Comments</u> Candidates were expected to provide any method of cooling the circuit, or preventing it heating in the first place. A wide variety of solutions were given and as long it is viable, it was credited.  Misconception Some candidates gave perfectly viable solutions, but these involved changes to the circuit, which was not allowed in the question. It is very important to make sure than any response does fit what is being asked.
			Total	8	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4		i	$12\,000 = \frac{Q}{4\pi\epsilon_0 r}$ $12\,000 = \frac{Q}{4\pi\epsilon_0 \times 0.19}$ $Q = 2.5(4) \times 10^{-7} \text{ (C)}$	C1 C1 A0	Allow $E = (V/d) = 6.316 \times 10^4$ C1 and $E = 6.316 \times 10^4 = \frac{Q}{4\pi\epsilon_0 \times 0.19^2}$ C1
		ii	1 $t = 78 \times 3600$ $(I =) \frac{2.5 \times 10^{-7}}{78 \times 3600}$ $I = 8.9 \times 10^{-13} \text{ (A)}$ 2 $(R =) \frac{6000}{9.0 \times 10^{-13}} \text{ or } 6.7 \times 10^{15} \text{ (}\Omega\text{) or } V =$ $IR \text{ and } R = \frac{\rho L}{A}$ $\frac{6000}{9.0 \times 10^{-13}} = \frac{\rho \times 0.38}{1.1 \times 10^{-4}}$ $\rho = 1.9 \times 10^{12} \text{ (}\Omega \text{ m)}$	C1 C1 A0 C1 C1 A1	There is no ECF from (b)(i) Note 2.54×10^{-7} gives an answer $9.0 \times 10^{-13} \text{ A}$ There is no ECF from (b)(ii)1 Take 12000 V as TE for this C1 mark, then ECF Note $8.9 \times 10^{-13} \text{ (A)}$ gives an answer $2.0 \times 10^{12} \text{ (}\Omega \text{ m)}$
			Total	7	
5		c		1	Examiner's Comments This was a well-answered question with most candidates correctly recalling that charge is conserved according to Kirchhoff's first law. A significant number of candidates distracted towards B; perhaps because of the unit of charge is the coulomb.
			Total	1	

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
6	B	1	Examiner's Comments This was not an easy question, but most candidates did extremely well in this multi-step calculation. The directions of the currents are important. The current at Y and the current in the resistor, must add up to 0.060 A. The charge passing point Y in a time of 10 s can be calculated using $\Delta Q = I\Delta t$ and finally the number of electrons can be determined by dividing the charge passing through Y by the elementary charge e. Therefore $\text{number of electrons} = \frac{(0.060 - 0.020) \times 10}{1.60 \times 10^{-19}} = 2.50 \times 10^{18}$ The key, correct answer, is B. The most popular distractor was C, in which a current of 0.060 A is assumed; the rest of the distractors were equally favourable. A was the answer for a current of 0.020 A and D was the answer for a current of 0.080 A. It is worth mentioning that most of the candidates just wrote down the correct answer in the box, without any calculations. On the surface, this looks reckless, but it is an excellent strategy if the numbers are being punched into the calculator correctly. The exemplar 1 below shows a typical correct answer, with numbers jotted down for visual help. Exemplar 1 Part of an electric circuit is shown below.  The direction of all the currents and the magnitude of two currents are shown. How many electrons pass through the point Y in 10 s? A 1.25×10^{18} B 2.50×10^{18} C 3.75×10^{18} D 5.00×10^{18} Your answer B [1] The candidate has the correct current at Y. Without the 0.040 A, the answer would have led to one of the distractors. It is worth pointing out that the final step of the calculation must have been done directly on the calculator. An excellent time-saving approach.

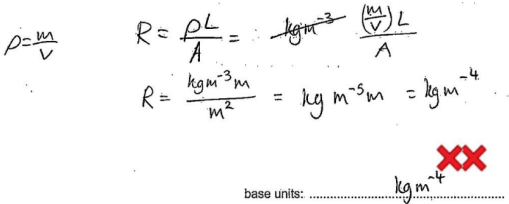
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			Total	1	

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
7	$(R = \frac{V}{I} = \frac{W}{QI}; Q = It)$ charge $\rightarrow$ A s or energy $\rightarrow$ kg m s⁻² $\times$ m or kg m² s⁻² (base units) kg m² A⁻² s⁻³	C1 A1	Allow other correct methods Allow Q or C or coulomb for 'charge'; E or W or joule or J or work done for 'energy' Allow 1 mark for J s⁻¹ A⁻² Allow $\frac{\text{kg m}^2}{\text{A}^2 \text{s}^3}$ or kg m² / (A² s³) Not kg m² / A² / s³ or kg m² / s³ / A² Examiner's Comments This was a challenging question, which provided the ideal opportunity for top-end candidates to use a variety of methods to get the correct S.I. base units of kg m² A⁻² s⁻³ for resistance. A significant number of candidates secured 1 mark for a partial answer with either charge $\rightarrow$ A s, or energy $\rightarrow$ kg m² s⁻². The rules for exponents were a bit perplexing for the low-scoring candidates. Many also misunderstood S.I. units. Exemplar 4 Derive the S.I. base units for resistance. $V = IR \quad R = \frac{V}{I} \quad V = \frac{W}{Q} = \frac{\text{Joules}}{\text{Coulombs}} = \frac{\text{kg m}^2 \text{s}^{-2}}{\text{A s}}$ $R = \frac{\text{kg m}^2 \text{s}^{-2}}{\text{A s}} \div \text{A} = \text{kg m}^2 \text{A}^{-2} \text{s}^{-3}$ base units: <u>kg m² A⁻² s⁻³</u> Two green checkmarks This exemplar illustrates a flawless answer from a top-end candidate. The equations are clear to see and follow. The units of each physical quantities are clearly identified and the appropriate S.I. units for the quantities have been successfully manipulated to give the correct answer. Compare this with the exemplar below which illustrates a common misconception.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					Exemplar 5  This exemplar illustrates a common error made by some candidates across the ability spectrum.  The resistivity ρ in the equation for resistance has been mistaken for density (which unfortunately has the same label). There can be no credit for wrong physics. It is vital to know your equations. Key:  Misconception
			Total	2	
8			A	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
9			B	1	Examiner's Comment All questions showed a positive discrimination, and the less able candidates could access the easier questions. The questions in Section A require careful reading and execution. Underlining or circling key information may help. Candidates are reminded not to use highlighter pens for this purpose. There is ample space for jotting down ideas and key equation, but it is best to do calculations on calculators to save time. Questions 1, 3, 10 and 14 proved to be particularly straightforward, allowing most of the candidates to demonstrate their knowledge and understanding of physics. At the other end of the scale, Questions 5, 9, and 15 proved to be more challenging. - Question 5 was on the superposition of waves and the relationship $\text{intensity} \propto \text{amplitude}^2$. The amplitude of the resultant wave is $0.4a$ and therefore the intensity of resultant wave must be $0.16I$. The most popular distractors were B and C. Less than half of the candidates got the correct answer of A. - Question 9 was about doubling the separation between two oppositely charged parallel plates. The only correct statement is D. Electric field strength is p.d. divided by the distance between the plates. Since both quantities double, the electric field strength, must remain the same. - Question 15 was about refraction and the equation $n \sin \eta = \text{constant}$ at the boundary between two materials. The ratio $n_1/n_2 = \sin 80^\circ / \sin 90^\circ = 0.98$; the correct is B. The most popular distractor was C, which was the inverse of the correct answer.
			Total	1	
10			C	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
11			B	1	
			Total	1	
12			$Q = It$ and $e = 1.6 \times 10^{-19}$ (C) number of electrons = $0.24 \times 10^{-6} \times 5.0 / 1.6 \times 10^{-19}$ number of electrons = 7.5×10^{12}	C1 C1 A1	
			Total	3	